Rehani Shinuka Perera



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Summary

I am a scientist with a background spanning biochemistry, molecular biology and biophysics, with hands-on experience in membrane protein purification, membrane-protein reconstitution, spectroscopy, mammalian cell culture and fluorescence microscopy. Experienced in independently developing and optimizing experimental workflows, troubleshooting complex scientific problems, analyzing experimental data and communicating scientific results to multidisciplinary audiences.

Technical Skills

Protein purification | Protein analysis | Reconstitution of proteins to membrane mimetics | cwEPR spectroscopy | Molecular cloning | Fluorescence microscopy | Mammalian cell culture | DLS spectroscopy | Data analysis and visualization | Experimental design and optimization | Peer-reviewed scientific publishing

Work Experience

Research Assistant and PhD candidate *Homburg (Saar), Germany*
Saarland University *Aug 2021 – April 2025*

- Conducting research for the doctoral thesis in the research group of Prof. Dr. Bianca Schrul.
- Co-first author on a multidisciplinary research project published in *Nature Communications*. doi.org/10.1038/s41467-024-48843-8
- Purified UBXD8 protein and reconstituted it into bilayer and monolayer membrane mimetics.
- Designed and optimized experimental workflows for generating artificial lipid droplets.
- Characterized UBXD8 using cwEPR spectroscopy.
- Performed mammalian cell culture and FRET confocal microscopy to investigate protein behavior in cellular systems.
- Proctored examinations for undergraduate medical students.

Research Assistant *Oxford, Ohio, USA*
Miami University *Jun 2019 – May 2020*

- Conducted research investigating the structural dynamics and topology of the membrane protein Holin-S105 for the master's thesis in the research group of Dr. Gary Lorigan.
- Co-author of a research paper including work from the master's thesis. doi.org/10.1016/j.bbamem.2022.184083
- Purified Holin-S105 and reconstituted it into membrane mimetics.
- Characterized Holin-S105 structure using cwEPR spectroscopy.
- Managed consumables and reagent orders for the Lorigan-biochemistry laboratory.

Teaching Assistant *Oxford, Ohio, USA*
Miami University *Aug 2018 – May 2019*

- Graduate teaching assistant for first-year undergraduate chemistry courses: CHM 144, CHM 145, CHM 111.
- Proctored examinations for undergraduate students.

Junior Executive - Business Development *Colombo, Sri Lanka*
Institute of Lean Management *Feb 2018 – Jun 2018*

- Supported senior management in planning and coordinating seminars, conferences and professional courses.
- Coordinated communication with external resource personnel and stakeholders.
- Served as primary coordinator for the international conference Lean Transformation Summit 2018 in Colombo, Sri Lanka.

Education

Doctorate in Natural Sciences *Aug 2021 – July 2026*
Saarland University, Homburg (Saar), Germany

- Grade: Magna cum laude
- Dissertation: Determination of the intramembrane positioning of the UBXD8 protein in phospholipid bilayer membranes and lipid droplet monolayers by electron paramagnetic resonance (EPR) spectroscopy. View dissertation.

M.Sc. in Chemistry *Aug 2018 – Aug 2020*
Miami University, Oxford, Ohio, USA

- Cumulative GPA: 1.25 (German Scale)
- Thesis: Determining the structural dynamics and topology of canonical Holin-S105 using cwEPR spectroscopy. View thesis.

B.Sc. (Hons) in Molecular Biology and Biochemistry *Mar 2013 – Mar 2017*
University of Colombo, Sri Lanka

- Cumulative GPA: 2.31 (German Scale)
- Thesis: Development of an RNA interference assay for microfilaria in *Culex quinquefasciatus* to study the functional role of novel parasitic nematode-specific protein (SdNP) of goat and sheep parasite *Setaria digitata*.

Selected Scientific Presentations

- Oral Presentation: Summer Camp of the SFB1027's IRTG 2024. *Bad Muenster am Stein, Germany.*
- Oral Presentation: EMBO Workshop- Lipid droplets:Metabolic hubs in health and disease 2023. *Sant Feliu de Guíxols, Spain.*
- Poster presentation: Cell Physics Conference 2023. *Saarbruecken, Germany.*
- Poster presentation: Summer Camp of the SFB1027's IRTG 2022. *Bad Muenster am Stein, Germany.*
- Poster presentation: 64th Annual Meeting of the Biophysical Society 2020 *San Diego, California, USA.*
- Poster presentation: Miami University Graduate Research Forum 2019. *Oxford, Ohio, USA.*

Languages

- **English** - Professional working proficiency (TOEFL iBT: 108/120)
- **German** - A2- actively improving

Certifications

- Matlab and Data Analysis (Saarland University) | May 2022
- Introduction to ÄKTA™ chromatography systems and UNICORN™ 7.7 control software (Cytiva) | July 2022
- Scientific Writing and Publishing online course (Springer Nature) | Aug 2021
- QMB1- Quantitative Methods for Biology (MATLAB)(HarvardX) | May 2020

References

Prof. Dr. Bianca Schrul
Principal Investigator - Saarland University
Research Supervisor - Doctoral Thesis
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Prof. Dr. Christopher Kay
Principal Investigator - Saarland University
Collaborator - EPR Spectroscopy
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